

Analyzing Factors That Influence Retail Sales Performance

Audience Questions and Responses

1. Why does the model have a relatively low R^2 value?

The R^2 value is low because the model does not include several important variables that influence sales, such as pricing, promotions, and customer behavior. As a result, the model can only explain a small portion of the variability in sales outcomes.

2. What additional variables would improve the model's predictive accuracy?

Including variables such as product pricing, discount levels, promotional campaigns, customer purchase history, and seasonal indicators would improve the model's predictive performance.

3. How might seasonality influence the sales trends observed over time?

The time series analysis shows fluctuations in sales over time, which suggests that seasonality or external factors influence demand. These patterns can impact forecasting accuracy if not properly modeled.

4. Why does the Technology category outperform the other product categories?

The Technology category likely performs best due to higher-priced items and strong consumer demand. This results in greater total revenue compared to other categories.

5. What factors could explain the lower sales performance in the South region?

The South region may be underperforming due to lower demand, fewer transactions, regional market differences or lower demand levels. It may also reflect differences in customer demographics or purchasing behavior.

6. How can this analysis be used to support better business decision-making?

This analysis helps identify high-performing categories and regions, allowing businesses to allocate resources more effectively. It also highlights areas where performance improvements can be targeted.

7. Would a different modeling approach improve prediction results?

More advanced models such as decision trees, random forests, or gradient boosting could better capture complex relationships in the data and improve predictive accuracy.

8. How might customer behavior influence sales beyond basic segment classification?

Customer behavior is a critical factor in sales performance. Differences in purchasing patterns, preferences, and spending habits across segments significantly influence overall revenue.

9. Are there any potential biases or limitations within the dataset used?

The dataset may contain biases due to missing variables, uneven distribution across regions or categories, and limited representation of real-world factors such as promotions or pricing strategies.

10. How could this analysis be extended to forecast future sales trends?

This project could be extended by incorporating additional data sources, applying more advanced predictive models, and developing time-series forecasting approaches to better predict future sales trends.